Homework 4

From Prototype to Product: Agentic Cell Erasing

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Code, presentation and recordings:
github.com/Arenananyan/PhotonicsAI-Lab-Cell-Erasing

Task 1. Extend and demonstrate the hardware-integrated AI system

(a) Project objective and implemented extension

We developed a prototype that connects natural-language requests to calibrated visual measurements and supervised printer motion. A user can inspect colored marks, select a cell by size or spatial relationship, review a path and authorize one bounded operation. In this project, "cells" are blue and green marks on a paper board; they are not biological cells.

The Homework 4 extension packages the hardware-facing components as a browser product: a persistent conversation workspace, measured cell reports, board and cell images, task suggestions, structured relational requests, operator approval, and a queue for multiple targets. The emphasis is on turning separate device interfaces into an inspectable, usable workflow.

(b) System architecture

Component	Responsibility
Browser frontend	Armenian/English conversations, history and deletion, light/dark themes, mobile layout, images, clarification and path approval.
Python backend	Task state, measured selection, conditions, scope validation, bounded execution and evidence records.
Groq model roles	Structured task interpretation, measured-ID selection, suggestions and visual assessment. The model proposes; the host validates.
Scanner MCP	Camera acquisition, board calibration, color detection, XY coordinates, areas and saved-frame artifacts.
Printer MCP	Status queries, complete-path preview, single-use execution tokens and constrained SSH/Klipper dispatch.

The reusable interaction loop

Request → observe → measure/select → preview → human approval → execute → verify.

The printer and scanner use separate MCP clients. Android IP Webcam supplies a local image endpoint; an optional Blackmagic iPhone adapter provides camera controls and preview. Tool evidence is treated as data, not as instructions. Physical action remains separate from a model's interpretation of the request.

(c) Reasoning, measurement and operator control

Implemented command families

Request	Resulting behavior
Inspect or count	Read-only report of available measurements; blue and green cells are ranked separately.
Largest / smallest	Host checks the requested measured rank. Near-equal sizes within 5% prompt a numbered choice with available cell crops.
Spatial relation	Find the blue cell nearest to the largest green cell, or report the distance between two size-selected cells.
If / then	Evaluate one supported count/area comparison on fresh calibrated evidence. A false condition finishes without motion.
Clean all	Freeze the initial eligible target queue, validate its paths, request one batch approval and attempt each target once.
Conversation reference	A previous selected cell can be referenced in the same chat; current geometry is re-observed before preparing a move.

Geometry is calculated by the host

For two calibrated cell centers, the reported distance is

$$d_{XY} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{in millimeters.}$$

The LLM receives measured IDs and geometry; it does not invent XY coordinates or calculate the authoritative distance. Nearest candidates within 3 mm of each other in distance require clarification. The 3 mm association tolerance is an operating threshold, not a demonstrated positioning-accuracy guarantee.

Fixed motion contract

The configured observation HOME is $(209, 208, 100)$ mm, distinct from firmware homing. The standard wipe uses an approach at $Z = 100$ mm, configured contact height $Z = 14.5$ mm, a $Y + 10$ mm offset and one clockwise circle of radius 10 mm, then raises and returns HOME. Default speed is 20 mm/s; accepted task speeds are bounded to at most 30 mm/s and further constrained by verified firmware limits. These settings describe this setup and do not establish physical clearance.

- Fresh calibrated observations must stabilize before a target is accepted; association is checked again before dispatch.
- Complete paths, homing state, mesh/coordinate agreement, firmware bounds, start position and motion gates are validated. The operator reviews clearance.
- Preview tokens are short-lived and single-use. A transport failure with an uncertain movement outcome never triggers an automatic motion retry.
- Batch cleaning attempts each initial target once. Valid evidence of residue is reported; an unavailable assessor can produce an uncertain result without repeating that cell. Unusable camera evidence still pauses the workflow.

Image-based result assessment

The verifier compares the same target region before and after a wipe, including colored-area measurements and an image-model assessment. More than 10% measured remaining color prevents a “clean” result. The possible verdicts are *clean*, *residue* and *uncertain*; they are visual assessments, not proof of mechanical contact. Simulated movement is identified explicitly.

(d) Product interface

The website keeps successive requests in one conversation. New chats are explicit; history, deletion, clarification and approval share the same workspace. Suggested actions fill a draft rather than executing a tool. Board pictures and individual cell crops are tied to their captured frame.

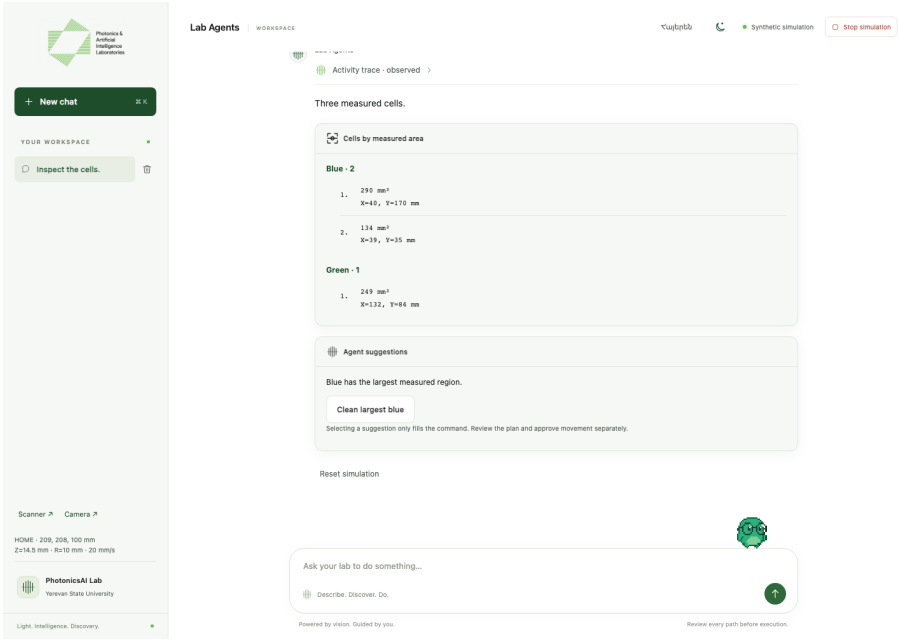


Figure 1. Desktop workspace in the light theme. This screenshot is from an offline UI test with simulated API data; it is not a live hardware result.

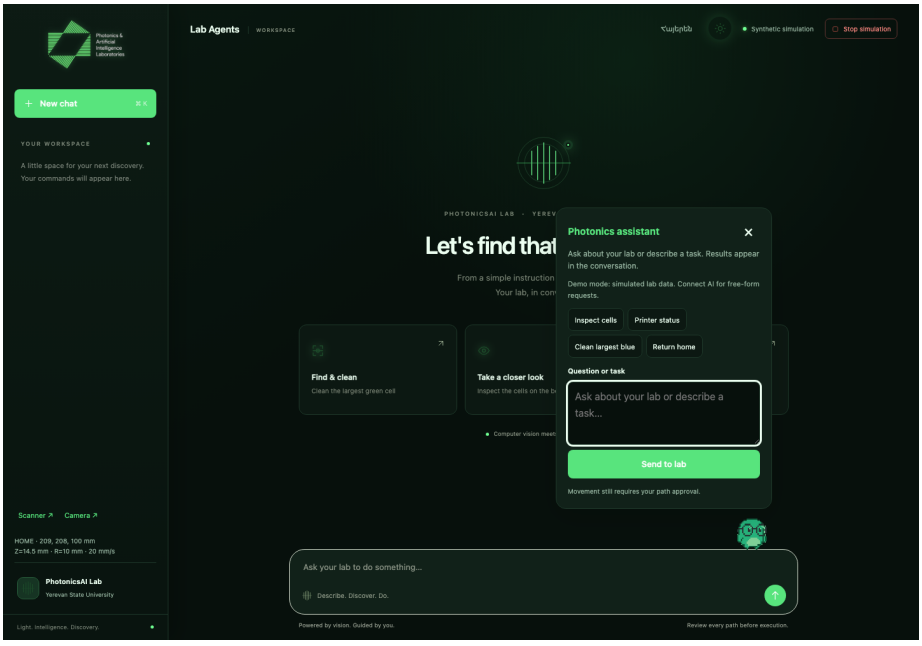


Figure 2. Dark theme and assistant panel. The panel enters the same conversation workflow; shortcuts do not bypass motion approval. Offline UI-test screenshot.

Local deployment and remote access

The application runs locally; a phone on the same network can access the dashboard when it is bound to the network interface. Remote browsers require the generated local login. The current HTTP deployment is intended for a trusted lab network; a production deployment would require stronger transport and access controls. Camera-specific calibration must be performed for the actual setup.

(e) Validation and supplied evidence

The submission copy was checked with the existing local Python environments. All listed software checks passed. Test fixtures, local sockets and mocked APIs were used; no real motion or live Groq request was issued during packaging.

Suite	Passed	Scope
Printer	143	Policy, mock state, local sockets, schemas and MCP
Scanner	61	Synthetic images, calibration, detection and camera-source API
Lab agents	104	Workflow, batch queue, spatial/size choices, evidence and access
Browser	29	Desktop/mobile layout, chat, themes, images and approval with mocked APIs
iPhone adapter	4	Viewport controls with synthetic JPEG input
Total	**341**	**Software checks; not a physical performance benchmark**

Hardware recordings

Two supplied recordings are included in the repository:

- `IMG_0030.MOV`: approximately 15.2 seconds.
- `IMG_7455.MOV`: approximately 9.8 seconds.

Sampled frames show the real printer and a board with colored marks. These clips provide hardware context; they do not by themselves establish the full request-to-approval chain, quantitative targeting accuracy, contact force, or successful ink removal. The evidence therefore supports an implemented software workflow and a physical prototype setup, rather than a claim of reliable autonomous erasing. Further controlled trials are needed to measure physical effectiveness.

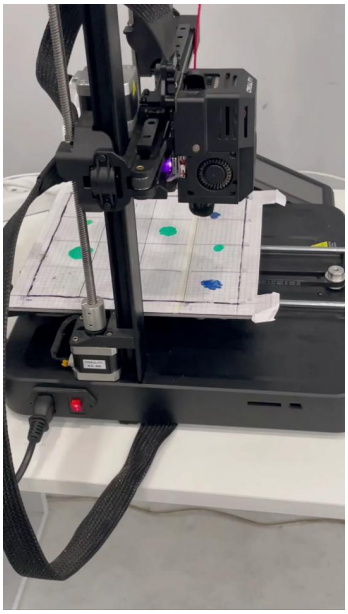


Figure 3. Sampled frame from the supplied `IMG_0030.MOV`. Real hardware context.

Reproducibility and artifact access

The repository preserves the four component folders, requirements, setup/check scripts, offline tests, website screenshots, two recordings and the supplied 10-slide presentation. API keys, SSH passwords, old run histories, camera calibration and virtual environments are excluded. A fresh network installation has not been independently verified.

From the repository root, install component environments with `python3 scripts/setup.py`. Run the offline Python suites separately with `python3 scripts/check.py`. The README documents mock demonstrations and operator-run real hardware startup. Scripted demo mode demonstrates the workflow, not live language-model reasoning.

Repository: PhotonicsAI-Lab-Cell-Erasing
Presentation: Agentic Cell Erasing System (10 slides)
Detailed checks: Package validation and evidence scope

(f) Limitations and future applications

Current prototype limits

Recognition is based on color/image processing and calibrated geometry, not a trained biological-cell or PCB-defect model. Lighting, occlusion, board movement and calibration affect measurements. Software tests do not quantify physical accuracy, removal rate or repeatability. A completed path does not prove that contact occurred or that ink was erased.

Live language interpretation and visual assessment depend on Groq availability. Minute/day quotas were encountered during development; the UI now exposes quota details and Retry-After when supplied. Follow-up references and complex commands remain bounded by the supported schema. The product is a supervised local lab prototype, not a general autonomous robot.

A reusable platform beyond cell erasing

The main transferable contribution is the interaction loop: interpret a request, locate a target, calculate geometry, propose a bounded action, obtain operator approval, execute and inspect the result. Colored-mark erasing is the first application used to explore that loop. The following are proposed development directions, not demonstrated capabilities of the submitted system.

Direction	Possible extension	Required development
Biomedical laboratories	Select a specimen region; position a camera or measurement probe under operator supervision.	Microscopy-specific recognition, finer positioning, sample-compatible tooling and application-specific validation. No clinical use is demonstrated.
PCB inspection and repair support	Locate a component or suspected defect; position a probe or assist a local cleaning operation.	PCB-specific perception, tool calibration, electrical precautions and force/temperature control where relevant. Actual board repair is not demonstrated.
Industrial inspection	Measure, mark or locally process a selected surface region.	Material-specific sensing and tooling, defined tolerances and validated before/after criteria.
Education and research	Supervised experiments and extensible instrument workflows.	New MCP adapters, repeatable procedures and structured experiment records.

Practical next steps

1. Measure targeting error and repeatability against independently measured board coordinates under varied lighting and camera placement.
2. Improve the end effector, establish contact/force feedback and evaluate removal with controlled before/after trials and predefined acceptance criteria.
3. Reduce model-call overhead and improve quota-aware waiting for interpretation; retain the prohibition on automatically retrying uncertain physical motion.
4. Adapt perception and tooling to one new domain, then validate that specific application before expanding the claims.

The prototype demonstrates how a hardware interface can become a conversational, measured and operator-controlled product. Its broader value is the extensible architecture; reliable task-specific physical performance is the next milestone.